and hence $\mathbb{L}(\boldsymbol{P}-1)=0$. Similarly, $\mu\mathbb{L}(\boldsymbol{P})=0$ since, as we have seen above, $\mathcal{M}$ and $\mathcal{M}\smallsetminus 0$ are in essential bijection. It so happens that these relations are the only ones needed to describe the kernels of $\mathbb{L}$ and $\mu\mathbb{L}$.

Theorem 3.22. *Suppose that $\mathbb{S}$ is (VF,Γ)-generated. For each $k\geq 0$ there is a canonical isomorphism of semigroups*

$$\int_+ : \mathbf{K}_+\,\mathrm{VF}[k] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[\leq k]/\mathrm{I}_{\mathrm{sp}}$$

such that $\int_+[A]=[\boldsymbol{U}]/\mathrm{I}_{\mathrm{sp}}$ if and only if $[A]=[\mathbb{L}\boldsymbol{U}]$. These isomorphisms are compatible with the inductive systems. Passing to the colimit of the groupifications, we obtain a canonical isomorphism of rings

$$\int : \mathbf{K}\,\mathrm{VF}_* \longrightarrow \mathbf{K}\,\mathrm{RV}[*]/(\boldsymbol{P}-1).$$

Similarly, for each $k\geq 0$ there is a canonical isomorphism of semigroups

$$\int_+^{\mu} : \mathbf{K}_+\,\mu\mathrm{VF}[k] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[k]/\mu\mathrm{I}_{\mathrm{sp}}$$

such that $\int_+^{\mu}[\boldsymbol{A}]=[\boldsymbol{U}]/\mu\mathrm{I}_{\mathrm{sp}}$ if and only if $[\boldsymbol{A}]=[\mu\mathbb{L}\boldsymbol{U}]$.

Referee R054 [Typo] – Measured integral notation.
Original source line 1386. The integral in this characterization should be $\int_+^{\mu}$, consistently with the immediately preceding display.

Taking the direct sum of the groupifications yields a canonical isomorphism of graded rings

$$\int^{\mu} : \mathbf{K}\,\mu\mathrm{VF}[*] \longrightarrow \mathbf{K}\,\mu\mathrm{RV}[*]/(\boldsymbol{P}).$$

This is a combination of two main theorems, Theorems 8.8 and 8.29, of [8].